



Short Film
Great Transitions: The Origin of Humans

hhmi | BioInteractive

Student Handout

NAME _____

DATE _____

This quiz was developed to supplement the viewing of the film *Great Transitions: The Origin of Humans*. Before watching the film, answer questions 1–3. As you watch, fill in question 4. Then answer the rest of the questions after watching the film.

Before Watching the Film

1. Which physical and behavioral characteristics do you think distinguish humans from other primates? List three key ones.

2. Charles Darwin hypothesized that humans and the African great apes all descended from a common ancestor. What evidence would you expect to find in the fossil record to support this hypothesis?

3. A “tool” is defined as “an instrument or device that is held by the hands and used to perform a particular task or function.” Think about the tools you have used so far today. What is the most primitive tool you have used today and why? What is the most advanced tool you have used and why?

As you watch

4. For each fossil discovery, write the name of the scientist(s) who found it, its approximate age and what combination of human characteristics it had.

Fossil	Scientist(s)	Fossil Age	Human Characteristic(s)
<i>Ardipithecus ramidus</i> (Ardi)			
<i>Australopithecus afarensis</i> (Lucy)			
Olduvai Hominid No. 7 (OH7)			
<i>Zinjanthropus</i>			



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After Watching the Film

5. At the beginning of the film, what are the three traits that Dr. Sean Carroll says make humans unique compared to our primate relatives?

6. What evidence was missing during Darwin's time that would have supported his hypothesis that humans evolved in Africa from ancestral primates?

7. What evidence has allowed scientists to conclude that the common ancestor of modern chimps and humans lived around 7 million years ago?

- Stone tool usage began showing up in the human lineage around 7 million years ago, and chimps do not use stone tools.
- From the changes scientists have observed in chimps over the past 200 years, they predict it will take at least 7 million years for today's chimps to evolve into more humans.
- Biological molecules such as proteins and DNA reveal differences between humans and chimps that would have taken around 7 million years to accumulate.
- Scientists know from whale evolution that it takes around 3.5 million years for brain size to double in volume in a species' lineage, and chimp brains are 300 cubic centimeters, while human brains are around 1,300 cubic centimeters.

8. Which of the following trends is illustrated by human ancestors over time?

- An increase in brain size
- A decrease in body size
- A reduction in the number of toes
- An increase in the size of the molars and the canine teeth

9. Which of the following is an anatomical feature that defines humans as bipedal hominids?

- A hairless face
- The absence of a prehensile tail
- Opposable digits on all four limbs
- A much shorter hip bone, broader front to back, that wraps around the side



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10. Which conclusion about human evolution can scientists infer from the existing *fossil* evidence?
- I. Humans and chimps shared a common ancestor around 7 million years ago.
 - II. Larger brains began evolving around 1.8 million years ago.
 - III. Humans began using tools at least 1.8 million years ago.
 - IV. Bipedality in hominids evolved at least 4.4 million years ago.
- a. I only
 - b. II and IV only
 - c. II, III, and IV only
 - d. I–IV
11. Which conclusion about human evolution can scientists currently infer from *molecular* (DNA and proteins) evidence?
- I. Humans and chimps shared a common ancestor around 7 million years ago.
 - II. Large brains began evolving around 1.8 million years ago.
 - III. Humans began using tools around 1.8 million years ago.
 - IV. Bipedality in hominids evolved at least 4.4 million years ago.
- a. I only
 - b. I and II only
 - c. II and IV only
 - d. I–IV
12. *Ardipithecus ramidus* (Ardi) is a hominid with a combination of chimplike and humanlike characteristics. Describe these characteristics.
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13. Before the discovery of Ardi, what kind of habitat did most paleobiologists hypothesize bipedality evolved in?
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14. What did the discovery of Ardi suggest about the kind of habitat in which bipedality evolved? How do we know?

15. How do scientists know how old the Ardi fossil is?

- a. The differences between Ardi's DNA and human DNA suggest that Ardi lived around 4.4 million years before modern humans.
- b. The Ardi fossil was found in sediment that was sandwiched between volcanic deposits that dated to 4.4 million years ago.
- c. Ardi's teeth reveal that she ate plants that went extinct around 4.4 million years ago.
- d. The stone tools found in the same sediment as Ardi date to 4.4 million years ago.

16. Explain how the Laetoli footprints found in Africa support the conclusion that *Australopithecus* was a biped.

17. What are the hallmarks of what Dr. White calls the third phase of human evolution? Check all that apply.

- Broadened diets _____
- Evolution of big teeth _____
- Evolution of bipedality _____
- Evolution of genus *Homo* _____
- Expansion out of Africa _____
- Expansion within Africa _____
- Reliance on culture _____
- Stone tool technology _____